

LEASEOS

Live Assist + Unified AI Assistant

Architecture and Build Plan

Version 2.0

19 September 2026

LeaseOS trucking / field operations platform

Purpose

Build one LeaseOS Assistant that can understand voice, LeaseOS context and explicitly shared Android screen content; help with paperwork, trucking, dispatch, mechanics, training and company operations; and carry out only authorized, typed actions through a default-deny authority gateway.

DESIGN STATUS: CURRENT PLANNING TRANCHE COMPLETE - BUILD PLAN READY

*This plan extends the existing LeaseOS AI Secretary, provenance, offline-first, role/permission and human-confirmation patterns.
It does not authorize autonomous bypass of compliance, safety, security, device or platform controls.*

Contents

- 1. Executive build brief
- 2. Locked design decisions
- 3. Non-negotiable safety and authority rules
- 4. Target architecture
- 5. Capability model and user experience
- 6. Android Live Assist architecture
- 7. Knowledge, memory, provenance and offline behavior
- 8. Data model and service contracts
- 9. Implementation phases and checkpoints
- 10. Test strategy and production gates
- 11. Security, privacy and platform compliance
- 12. Pilot and rollout
- 13. Recommended backlog
- 14. Source alignment and reference notes

- 15. Assistant Brain / Orchestrator build plan

- 16. Knowledge / RAG Gateway build plan

- 17. Multimodal Evidence Fabric build plan

- 18. Integrated implementation sequence and checkpoint gates

- 19. Current planning boundary and next subsystem

Recommended delivery strategy

Do not start with whole-device control. First build the authority gateway, typed action registry and unified AI Secretary. Then add read-only Live Assist. Only after the evidence, confirmation and policy gates are proven should LeaseOS enable deterministic cross-app actions.

1. Executive build brief

LeaseOS Live Assist is not a separate chatbot and not a generic remote-control bot. It is the visual/voice interaction layer for one unified LeaseOS Assistant. The user talks to one assistant; behind the scenes, an orchestrator routes work to approved capabilities for paperwork, driving, routing, dispatch, compliance, billing, training, mechanics and company operations.

The build must preserve a hard separation between AI reasoning and authoritative system writes. AI can observe, explain, extract, recommend and prepare. Actual business changes pass through a typed Action Gateway that checks tenant, identity, role, device trust, job context, vehicle state, risk, evidence and confirmation requirements before executing.

Primary outcome

A driver, mechanic, dispatcher or office user should be able to say what they need in ordinary language, optionally share the Android screen, and receive help that is grounded in LeaseOS records and approved sources - without giving an LLM direct database write access or unrestricted device control.

Success criteria

- One visible LeaseOS Assistant; specialist routing is internal.
- Works on personal/BYOD Android and company-managed fleet Android devices.
- Hybrid local-first processing: wake phrase, basic OCR/UI understanding and simple commands stay local when practical; cloud models are used only when needed.
- Parked / moving / emergency modes materially change allowed actions, not merely the UI.
- Important answers show source, freshness, confidence and whether content is measured, observed, historical, inferred or confirmed.
- No consequential action is committed without the required authority and confirmation.
- Offline operation degrades safely: cached data remains usable, stale/live dependencies are labeled, writes queue for synchronization.
- Raw screen/audio is transient by default; durable business facts are stored as structured records with provenance.
- Mechanics assistance can diagnose and prepare work, but cannot autonomously declare a vehicle safe or release it to service.

2. Locked design decisions

Decision	Approved direction
Screen scope	Both: LeaseOS-native context plus explicit whole-screen sharing sessions.
Action level	See + understand + operate, but only through typed/approved actions and confirmation gates.
Device deployment	Both BYOD/personal devices and company-managed fleet devices.
AI topology	Hybrid local-first with cloud escalation when required.
Driving safety	Parked / moving / emergency state model.
Assistant UX	One LeaseOS Assistant with automatic specialist routing.
Knowledge	Approved company knowledge + approved external/live sources with provenance.
Permissions	Secure LeaseOS defaults + company-customizable roles; non-bypassable safety floor.
Proactivity	Mixed proactive mode with policy-controlled interruptions and task inbox.
Invocation	Button + voice + optional local wake phrase such as “Hey LeaseOS”.

Decision	Approved direction
Memory	Layered memory/retention: session, job, user, equipment, company and audit.
Offline	Graceful degradation with local/cached capabilities and queued synchronization.
Android control	Native APIs/intents first; deterministic accessibility automation only where policy-permitted; no unrestricted autonomous LLM tap loop.

3. Non-negotiable safety and authority rules

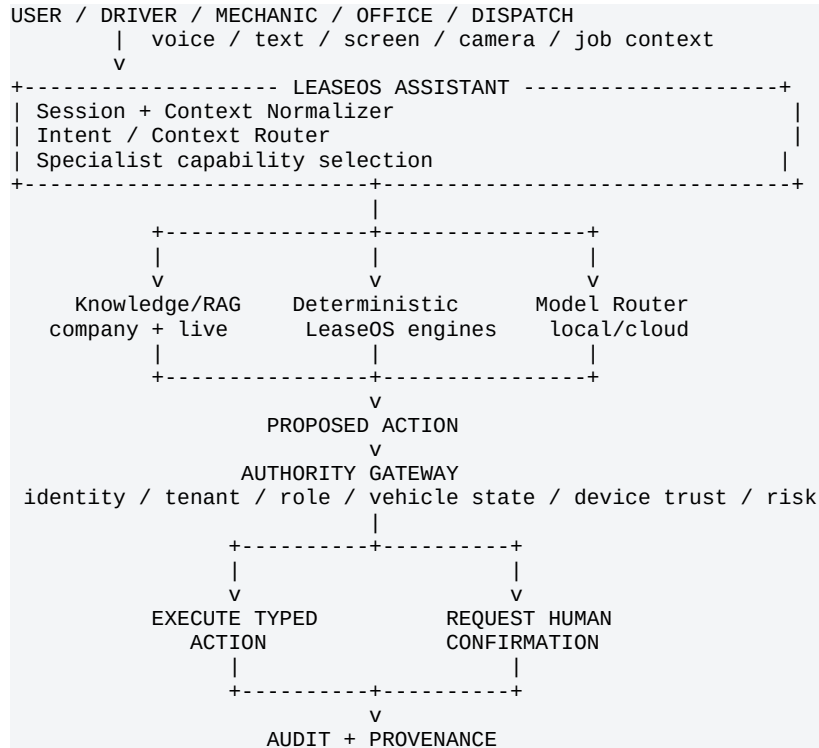
Rule	Implementation meaning
AI does not write directly to authoritative tables	All writes go through typed service/action contracts.
Unknown is not safe	Missing or stale evidence produces UNKNOWN/REVIEW, never a fabricated pass.
Proposal before commitment	OCR, voice extraction, predicted maintenance, billing lines and compliance changes remain proposed until required confirmation occurs.
User authority caps AI authority	The assistant cannot obtain more authority than the authenticated user and current delegated role.
Consequential actions step up	HOS corrections, signatures, billing submission, vehicle release, dispatch acceptance, deletions, external communications and permission changes require explicit policy checks and usually human confirmation.
Driving state is a control input	Moving state disables distracting and nonessential interaction pathways.
Protected content stays protected	LeaseOS does not attempt to bypass Android secure-screen, enterprise, biometric, privacy or app restrictions.
Evidence remains explainable	A user can ask “Why?” and see the records/sources, timestamps, confidence and inference boundary.
Tenant isolation is absolute	No cross-company data leakage through retrieval, chat history, screen observations, embeddings, logs or model context.
Raw capture is temporary by default	Store structured facts and evidence references rather than continuous recordings unless a defined evidence policy requires retention.

Action risk ladder

Level	Examples	Human confirmation
L0 - Observe	Read screen/context; explain; search approved knowledge.	No
L1 - Assist	Highlight, calculate, guide, draft.	No

Level	Examples	Human confirmation
L2 - Prepare	Populate forms, work-order drafts, dispatch/invoice drafts.	Usually no to save draft
L3 - Routine execute	Reversible, approved actions such as navigation handoff, filters, draft save, attach evidence.	Policy dependent
L4 - Consequential	HOS correction, billing submit, dispatch acceptance, safety declaration, vehicle release, signature, delete, external send.	Yes
L5 - Restricted	Bypass compliance/safety/security, fabricate evidence, impersonate signature, weaken controls, mark unknown as verified.	Never

4. Target architecture



Core logical modules

Module	Responsibility
Assistant Session Manager	Starts/ends assistant sessions; tracks user, device, job/unit, vehicle state, connectivity and allowed scopes.
Intent/Context Router	Routes a request to paperwork, routing, HOS, mechanics, training, dispatch, billing, company operations or a combination.
Capability Registry	Declares what each capability can read, prepare or execute; inputs, outputs, risk level, confirmation, offline availability.

Module	Responsibility
Authority Gateway	Default-deny enforcement point for every assistant-initiated action.
Action Registry	Typed action vocabulary such as OPEN_JOB, NAVIGATE_TO, CREATE_DEFECT, PREPARE_FIELD_TICKET, CHECK_HOS and CREATE_WORK_ORDER.
Knowledge Gateway	Retrieves approved company knowledge, LeaseOS records, authoritative regulatory/OEM material and live external data with source metadata.
Model Router	Chooses local rules/model vs cloud model based on task complexity, privacy, connectivity and policy.
Live Assist Capture Engine	Explicit Android MediaProjection session, frame-change detection, local OCR/UI parsing and redaction.
Platform Adapters	Android intents/deep links, 511/map integrations, email/messaging, OEM/telematics and other official APIs.
Task Inbox	Turns unresolved assistant work into durable operational tasks.
Audit/Provenance Ledger	Records source, confidence, actor, approver, action, before/after, device/session and result.

5. Capability model and user experience

5.1 One assistant, specialist capabilities

Capability	Scope
Paperwork	Voice/screen/camera extraction, gap detection, read-back, proposed forms, tickets, manifests, inspections and billing evidence.
Driver/Trucking	Job instructions, routes, offline maps, HOS assistance, inspections, load/stop context, road restrictions and contacts.
Dispatch	Assignment context, readiness, route/status explanation, job contacts, exceptions and queued decisions.
Mechanics	Defects, diagnostic context, work orders, unit history, sensor readings, parts/labor/evidence and guided troubleshooting.
Safety/Compliance	Explains blocks, missing evidence and rule state; never invents compliance or bypasses hard gates.
Billing/Office	Prepared invoices, evidence completeness, rejected/missing ticket detection, customer requirements and review queues.
Training/How-to	LeaseOS usage, company SOPs, approved training, equipment procedures and escalation to authoritative manuals.
Company Operations	Employee/unit/document status, task queues, expiries, client requirements, summaries and authorized administration.

5.2 Context-sensitive behavior

State	Allowed behavior
Parked	Full voice and visual assistance; prepare/review forms; routine approved actions; L4 confirmation as required.
Moving	Voice-first; navigation, route/HOS/safety information, read-aloud, alerts and task queuing. Complex typing/review and nonessential cross-app interaction disabled.
Emergency/Critical	Surface coordinates, emergency contacts/ERP/ERAP, critical instructions and approved communication shortcuts. Emergency mode does not bypass compliance/security controls.
Offline	Use cached map/company/manual/job data; local extraction/rules where available; label stale dependencies; queue sync/actions requiring connectivity.

5.3 Explainability

Operationally meaningful answers should support a “Why?” view. The answer payload should carry source class, source identifier, observed/verified time, freshness, confidence and the boundary between evidence and model inference.

```

Example diagnostic statement
MEASURED   coolant_temp = 112 C      source: ECU sensor
OBSERVED   residue near upper hose   source: mechanic photo
HISTORICAL similar loss 18 days ago   source: W0-1842
INFERRED    probable cooling leak      source: assistant reasoning
CONFIRMED   upper radiator hose leak  verified_by: mechanic

```

6. Android Live Assist architecture

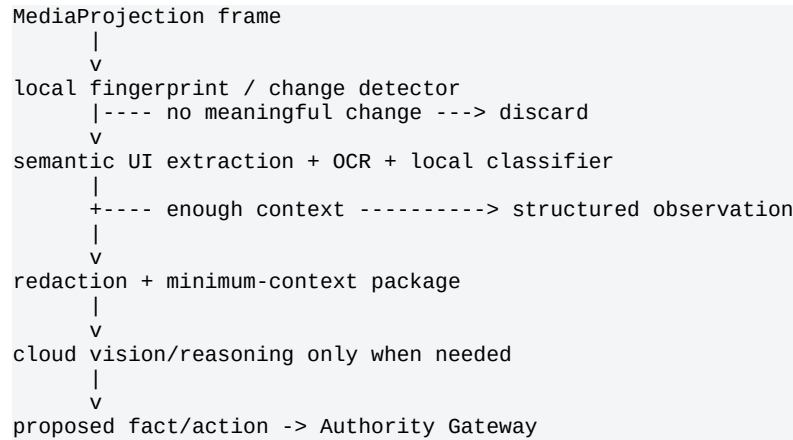
6.1 BYOD/personal devices

- Use Android MediaProjection for explicit whole-screen capture sessions. User consent is requested for each capture session where required by Android behavior.
- Show a persistent LeaseOS “LIVE ASSIST ACTIVE” control plus Android/system capture indication. Either stop path terminates capture and releases resources.
- Process frame fingerprints, UI semantics and OCR locally first; do not continuously stream video to the cloud.
- Redact sensitive regions when possible before cloud escalation. Treat password/PIN/biometric/secure screens as user-only steps.
- Use official APIs, LeaseOS APIs, Android intents and deep links before any accessibility-based automation.
- If AccessibilityService is used for automation, constrain it to declared, narrow, deterministic workflows. Do not give an LLM an unrestricted plan-and-tap loop.

6.2 Company-managed fleet devices

- Use Android Enterprise/MDM policy to standardize allowed apps, security posture, data-sharing policy and Live Assist availability.
- Respect device-owner/profile-owner restrictions that disable content capture or AI/agent automation.
- Keep company policy separate from the non-bypassable LeaseOS safety floor: MDM may reduce capability, never silently expand assistant authority.
- Support separate profiles/policies for driver tablets, shop/mechanic devices and office/dispatch devices.

6.3 Capture pipeline



6.4 Android/Google Play implementation constraints

Current platform constraint - re-verify before store submission

Google Play policy permits narrow, clearly understood deterministic accessibility automation, but prohibits using AccessibilityService so an app autonomously initiates, plans and executes actions or decisions. Build LeaseOS around typed actions, intents/APIs and deterministic adapters - not unrestricted AI UI control.

Area	Build requirement
MediaProjection	Request user consent for capture; use required foreground service type/permission; handle projection stop callback and release surfaces/resources.
AccessibilityService	Prominent disclosure/consent and Play declaration if used; narrowly scoped deterministic workflows; prefer narrower APIs whenever possible.
Secure/protected screens	Honor platform/app capture restrictions. Do not attempt bypasses.
Enterprise restrictions	Detect and honor device policy restrictions such as content-capture or AI automation controls.
Third-party AI	Data-safety/user-data responsibilities remain with LeaseOS; minimize data and disclose processing appropriately.

7. Knowledge, memory, provenance and offline behavior

7.1 Source classes

Source class	Use
AUTHORITATIVE_REGULATORY	Government/regulator rule or dataset.
OEM_TECHNICAL	Manufacturer/service documentation.
COMPANY_POLICY	Approved employer SOP, customer requirement or internal policy.
LEASEOS_RECORD	Operational record already in LeaseOS.

Source class	Use
LIVE_EXTERNAL	Current road/511/weather/partner API response.
AI_GENERAL_KNOWLEDGE	Explanation only; never sufficient evidence for a consequential operational fact.

7.2 Layered memory

Layer	Retention/use
Session	Temporary conversation/Live Assist context; short TTL by default.
Job	Current assignment, customer, route, unit/trailer, forms, contacts, evidence and unresolved tasks.
User	Approved role, accessibility, units/preferences and non-sensitive workflow preferences.
Equipment	Defects, maintenance, sensor events, inspections, parts and recurring issues.
Company	Approved SOPs, forms, training, customer rules, rates and knowledge.
Audit	Immutable evidence of consequential actions, approvals and outcomes; not ordinary chatbot memory.

7.3 Offline-first behavior

Connectivity state	Behavior
FULL	Local + cloud AI, live sources, online LeaseOS services.
LIMITED	Local AI/rules + cached LeaseOS data; cloud/live work queued or selectively retried.
OFFLINE	Local voice commands where supported, cached knowledge/maps/manuals, paperwork drafting, photo/note evidence, queued sync.
STALE/UNKNOWN	Anything requiring fresh external verification is clearly marked and cannot silently satisfy a compliance/safety gate.

8. Data model and service contracts

Names below are recommended logical contracts. Adapt them to the existing LeaseOS schema and naming conventions instead of creating parallel concepts.

8.1 Recommended tables/entities

Entity	Key content
assistantSessions	tenantId, userId, deviceId, jobId/unitId, startedAt, endedAt, mode, drivingState, scopes, connectivityState

Entity	Key content
assistantTasks	type, context refs, owner/role, priority, status, dueAt, sourceSessionId, blockingReason
assistantActions	actionType, argsHash, actor, actingRole, riskLevel, state, result, reversible, timestamps
assistantApprovals	actionId, requiredRole, approver, method, decision, reason, timestamp
assistantEvidenceRefs	action/task/message ref -> source class/id/version, observedAt, confidence, freshness, excerpt/hash
liveAssistSessions	projection/session state, allowed capture scopes, retention policy, start/stop reason, policy snapshot
screenObservations	ephemeral/short-lived structured observation; sourceApp, UI element/text, confidence, frame hash, redaction state
capabilityDefinitions	capability/action metadata: input schema, output schema, risk, offline support, required scopes, confirmation
assistantPolicies	tenant overrides within LeaseOS hard limits: proactivity, retention, allowed adapters, cloud escalation, role mappings
knowledgeSources	source identity, authority class, version, jurisdiction/customer/OEM scope, verifiedAt, license/usage constraints
assistantAuditEvents	append-only assistant lifecycle/action/approval/security events linked to existing audit ledger

8.2 Provenance type

```
Provenance {
  sourceClass, sourceId, sourceVersion,
  observedAt, verifiedAt, verifiedBy,
  confidence, freshness, status,
  precision?, evidenceHash?
}
```

status: pending | confirmed | rejected | unknown | stale

8.3 Typed action contract

```
ActionDefinition {
  id: "CREATE_WORK_ORDER",
  inputSchema,
  allowedRoles,
  allowedDrivingStates,
  riskLevel: "L2",
  confirmationPolicy,
  requiredEvidence,
  reversible,
  offlinePolicy,
  executorAdapter
}
```

8.4 Initial action vocabulary

- OPEN_JOB, OPEN_ROUTE, OPEN_UNIT, SHOW_DOCUMENT, CALL_JOB_CONTACT
- NAVIGATE_TO, CHECK_ROUTE, CHECK_HOS, CHECK_DISPATCH_READINESS
- CREATE_DEFECT, CREATE_WORK_ORDER, ADD_WORK_ORDER_EVIDENCE, PREPARE_UNIT_RELEASE
- ADD_LOAD, ADD_UNLOAD, ADD_DISPOSAL_TICKET, ATTACH_PHOTO, ATTACH_DOCUMENT

- PREPARE_FIELD_TICKET, PREPARE_MANIFEST, PREPARE_BILLING, REQUEST_SIGNATURE
- OPEN_ERP, SHOW_EMERGENCY_CONTACTS, QUEUE_TASK, EXPLAIN_BLOCK, SHOW_WHY

9. Implementation phases and checkpoints

Sequencing rule

Build authority, provenance and typed actions before adding screen control. Build read-only screen understanding before cross-app automation. Treat every checkpoint as a shippable safety boundary.

P0 - Baseline and contracts

Scope	Deliverables	Exit gate
Inventory existing AI Secretary, audit, role, offline, mobile and routing contracts; freeze duplicate concepts; define module ownership and feature flags.	Architecture map; ADRs; dependency map; feature flags; policy snapshot tests.	No new AI write path is allowed until P1.

P1 - Authority Gateway + Action Registry

Scope	Deliverables	Exit gate
Implement default-deny action evaluation, L0-L5 risk, role/tenant/device/job/driving-state checks, approvals and audit receipt.	Typed action registry; authorization evaluator; approval workflow; audit schema.	Every assistant write must fail closed when policy context is missing.

P2 - Unified AI Secretary / Orchestrator

Scope	Deliverables	Exit gate
Unify paperwork and specialist routing behind one session/orchestrator; schema-constrained extraction; minimum-question gap detection; read-back.	Session API; intent router; capability registry; structured proposals.	No direct model-to-DB mutation; all proposed fields carry provenance.

P3 - Knowledge + Provenance Gateway

Scope	Deliverables	Exit gate
Build source classes, retrieval filters, freshness/authority metadata, "Why?" payloads and tenant isolation.	KnowledgeSource registry; provenance adapter; source/freshness UI contract.	Important answer can enumerate evidence and inference boundary.

P4 - Voice + driving-state shell

Scope	Deliverables	Exit gate
Button/mic flows, optional local wake phrase, parked/moving/emergency behavior, visible mic/session controls.	Voice session manager; driving-state policy hook; command queue.	Moving-mode test suite proves distracting actions are unavailable.

P5 - In-app Live Assist

Scope	Deliverables	Exit gate
Assistant understands LeaseOS UI state directly without whole-screen capture; highlights and guides users; opens typed LeaseOS actions.	Screen/context contract inside LeaseOS; help overlays; how-to routing.	Can complete top LeaseOS help workflows without MediaProjection.

P6 - Whole-screen read-only Live Assist

Scope	Deliverables	Exit gate
Add MediaProjection session, local frame-change detection, OCR/UI parsing, redaction and cloud escalation. No cross-app execution yet.	Android capture service; observation pipeline; session permission UI; stop lifecycle.	Capture always explicit/visible; revocation instantly stops processing; raw frames obey retention.

P7 - Deterministic external action adapters

Scope	Deliverables	Exit gate
Add native intents/deep links/APIs and tightly scoped deterministic accessibility actions only where needed.	Adapter SDK; Maps/navigation adapter; approved messaging/email handoff; selected app workflows.	No arbitrary tap loop; each action maps to a reviewed ActionDefinition.

P8 - Hybrid model router + offline assistant

Scope	Deliverables	Exit gate
Local/simple task routing, cloud escalation policy, cached knowledge/manuals, queued assistant work and sync conflict handling.	Model router; offline capability manifest; encrypted queue; stale-state UX.	Loss of network does not convert unknown data into verified results.

P9 - Managed fleet mode

Scope	Deliverables	Exit gate
Android Enterprise/MDM policy integration, device posture and managed restrictions.	Managed policy reader; fleet admin settings; device compliance indicators.	Enterprise restrictions always reduce/disable capability safely.

P10 - Mechanic + company operations expansion

Scope	Deliverables	Exit gate
Mechanic diagnostics/work orders plus office/dispatch/billing/training capabilities using the same contracts.	Mechanic context resolver; equipment evidence; operations capability modules.	Mechanic assistant never independently closes safety-critical release decision.

P11 - Proactive assistant + task inbox

Scope	Deliverables	Exit gate
Policy-controlled alerts, missing-evidence detection, billing-ready queue, expiring docs, defects and unresolved assistant tasks.	Task rules; notification policy; task inbox UI; escalation.	No alert can auto-escalate authority or auto-commit L4 actions.

P12 - Security/privacy/policy hardening

Scope	Deliverables	Exit gate
Threat model, Play declarations, disclosure/consent, data minimization, log redaction, abuse tests, retention controls.	Policy documentation; privacy UX; pen-test findings; store submission checklist.	Zero known critical security/privacy issues; declarations match actual behavior.

P13 - Field pilot and production gate

Scope	Deliverables	Exit gate
Real-device testing across supported Android versions, weak/offline networks, personal and managed devices, driver/shop/office pilots.	Pilot report; telemetry; defect burn-down; rollback plan; release runbook.	Production gates in Section 10 all green.

Recommended release checkpoints

Checkpoint	Phases	Capability
Checkpoint A - Assistant Core	P0-P3	One assistant, provenance, typed actions, no screen sharing.
Checkpoint B - Voice/In-app	P4-P5	Voice + in-app guided assistance, driving-state controls.
Checkpoint C - Live Assist Observe	P6	Whole-screen understanding only; no autonomous cross-app execution.
Checkpoint D - Controlled Actions	P7-P9	Approved adapters, offline/hybrid behavior and managed fleet mode.
Checkpoint E - Operational Suite	P10-P11	Mechanics/company operations/proactive tasks.
Production Candidate	P12-P13	Security/policy/device/pilot gates complete.

10. Test strategy and production gates

10.1 Required automated test families

- Authorization matrix: every action x role x tenant x driving state x risk level.
- Tenant isolation: retrieval, embeddings, screen observations, task queues and audit refs.
- Proposal/confirmation: uncertain OCR/voice values never silently become confirmed records.
- Capture lifecycle: deny, start, rotate, app switch, protected content, revoke, process death, reboot.
- Moving-mode enforcement: prohibited UI/action pathways remain unavailable even if requested by model output.
- Offline/stale handling: cached data labels, queued writes, retries, conflict resolution and no false verification.
- Action idempotency/replay: duplicate model/tool calls cannot duplicate invoices, tickets, messages or work orders.
- Model failure/hallucination: invalid/unknown action or argument is rejected before executor.
- Approval step-up: L4 action cannot execute with missing/expired/wrong-role approval.
- Managed policy: content-capture/AI restrictions are detected and honored.
- Retention/redaction: transient screen frames/audio are removed according to policy and sensitive values are not leaked to logs.
- Accessibility and one-handed/mobile: large text, screen reader, gloves/large touch targets where supported, voice fallback.

10.2 Production gates

Gate	Production requirement
G1	0 direct LLM writes to authoritative LeaseOS business tables.
G2	100% assistant actions map to registered typed ActionDefinitions.
G3	100% L4 actions show required confirmation and audit receipt.
G4	Unknown/stale evidence cannot produce PASS/SAFE/VERIFIED.
G5	All important operational answers expose source + time/freshness + confidence/inference status.
G6	Capture sessions require explicit current consent/visibility and stop cleanly on revocation.
G7	No unrestricted autonomous AccessibilityService planning/execution path exists in Play-distributed build.
G8	Tenant isolation and authorization tests pass for every assistant data path.
G9	Offline behavior is safe and understandable; queued records are durable and encrypted.
G10	Security/privacy review has no unresolved critical/high issues affecting production use.
G11	Field pilot covers personal + managed Android, moving/parked states, weak network and real paperwork/mechanic workflows.

Gate	Production requirement
G12	Rollback/kill switch can disable Live Assist or specific adapters without disabling core LeaseOS.

11. Security, privacy and platform compliance

11.1 Threat model priorities

Threat	Mitigation
Prompt injection from screen/doc	Treat external content as untrusted data. It cannot redefine assistant policy or authorize actions.
Cross-tenant retrieval	Tenant-scoped indexes/queries and policy-enforced joins; tests for every retrieval path.
Credential capture	Detect/redact sensitive UI where possible; never persist passwords/PINs/tokens from capture; secure screens remain user-only.
Over-broad screen retention	Frame TTL and structured-observation model; evidence retention must be explicit and job/policy linked.
Unauthorized external action	Typed allowlist + risk/role/context checks + idempotency + confirmation.
Model tool abuse	Model can request only known capabilities; gateway validates all arguments server-side.
Lost/stolen device	Device binding, secure local storage, session timeout, revocation, managed wipe where applicable.
Audit tampering	Append-only audit trail with integrity controls and durable references to evidence/action receipts.
Proactive distraction	Moving-mode suppression and policy-controlled interrupt classes.
Stale regulatory/live data	Freshness thresholds and explicit UNKNOWN/STALE state; no silent fallback to general model knowledge.

11.2 Privacy defaults

- Whole-screen sharing is opt-in, session-scoped and visibly active.
- Raw screen/video/audio is not a general historical archive.
- Cloud escalation transmits the minimum context needed for the task and records source/processing policy.
- Company administrators can shorten optional AI-memory retention but cannot erase legally/operationally required audit records through the conversational assistant.
- User-facing permission dashboard shows current screen, mic, location, cloud and external-action access.
- Assistant logs must avoid secrets and unnecessary personal content.

12. Pilot and rollout

12.1 Recommended pilot cohorts

Cohort	Size	Focus
Driver pilot	5-10 drivers	Voice, parked/moving controls, paperwork preparation, in-app help, read-only Live Assist, route handoff.
Shop pilot	2-5 mechanics	Defect/work-order flows, photos, unit history, OEM/company knowledge, diagnostic explanations.
Office/dispatch pilot	3-5 users	Billing readiness, missing evidence, dispatch exceptions, task inbox, role/approval behavior.
Managed-device pilot	Small fleet subset	MDM policy, capture restrictions, device trust, remote policy change and rollback.

12.2 Pilot metrics

- Paperwork completion time and number of fields/questions eliminated.
- Assistant proposal acceptance/rejection rate by field and workflow.
- False-positive/false-confidence incidents, especially safety/compliance/mechanics.
- Number of times moving-mode correctly deferred a distracting/consequential task.
- Cloud calls, bandwidth, battery impact and local-vs-cloud routing ratio.
- Capture-session abandon/revoke rate and privacy complaints.
- Missing-evidence reduction before billing/dispatch.
- Mechanic diagnostic usefulness without increasing unsafe release behavior.
- Offline queue success, sync conflict rate and time-to-recovery.
- Support requests resolved by LeaseOS How-To without human escalation.

13. Recommended backlog

ID	Area	Task	Phase
LA-001	Architecture	Map existing AI/role/audit/offline modules; identify reuse and duplicates.	P0
LA-002	Contracts	Create shared Provenance contract and status vocabulary.	P0
LA-003	Security	Create AssistantAction + Approval receipt contract.	P1
LA-004	Security	Implement default-deny Authority Gateway.	P1
LA-005	Actions	Build Action Registry with first 15 LeaseOS actions.	P1
LA-006	AI	Create unified assistant session/orchestrator.	P2

ID	Area	Task	Phase
LA-007	AI	Connect schema-driven paperwork engine and read-back.	P2
LA-008	Knowledge	Implement source-class/freshness/provenance retrieval.	P3
LA-009	UX	Build “Why?” evidence panel contract.	P3
LA-010	Voice	Add push-to-talk + optional local wake phrase.	P4
LA-011	Safety	Implement driving-state action matrix.	P4
LA-012	UX	Build in-app Assist overlay/help mode.	P5
LA-013	Android	Implement MediaProjection service + explicit consent lifecycle.	P6
LA-014	Android	Implement frame fingerprint/OCR/redaction pipeline.	P6
LA-015	Android	Build Live Assist permission/session dashboard.	P6
LA-016	Adapters	Create adapter SDK and navigation intent adapter.	P7
LA-017	Adapters	Add deterministic accessibility runner only for reviewed narrow workflows.	P7
LA-018	Offline	Build model router, local capability manifest and queued assistant actions.	P8
LA-019	Enterprise	Read/enforce Android Enterprise restrictions and device policy.	P9
LA-020	Mechanics	Build unit/defect/work-order mechanic context resolver.	P10
LA-021	Ops	Add billing/dispatch/training/company-operation capability definitions.	P10
LA-022	Tasks	Build assistant task inbox + proactive rule engine.	P11
LA-023	Privacy	Implement retention/redaction policy tests and disclosure UX.	P12
LA-024	Compliance	Prepare Play Accessibility/MediaProjection/Data Safety submission evidence as applicable.	P12

ID	Area	Task	Phase
LA-025	QA	Real-device matrix + pilot runbook + rollback/kill switch.	P13

14. Source alignment and reference notes

14.1 Existing LeaseOS alignment

This plan intentionally extends, rather than replaces, existing LeaseOS patterns found in the project material:

- LEASEOS_BUILD_PLAN-3: a single schema-driven conversational form engine; per-field value/precision/source/confidence; minimum missing questions; mandatory read-back; commit with provenance; automation proposes and humans confirm before authoritative commitment.
- LeaseOS prototype role model: Driver, Dispatcher, Mechanic, Office, Manager and Administrator with distinct visibility/change rights.
- LeaseOS offline prototype: encrypted local storage, queued events/photos and explicit last-sync/offline status.
- LeaseOS routing evidence design: unknown is not treated as clear; decisions retain evidence sufficient to reproduce/explain the verdict.

14.2 Android / Google Play references checked for this plan

Reference	URL
Google Play - Use of the AccessibilityService API	https://support.google.com/googleplay/android-developer/answer/10964491?hl=en
Android Developers - Foreground service types / mediaProjection	https://developer.android.com/develop/background-work/services/fgs/service-types
Android 14 behavior - user consent for each MediaProjection capture session	https://developer.android.com/about/versions/14/behavior-changes-14
Android media capture / MediaProjection	https://developer.android.com/media/platform/av-capture
Android UserManager - DISALLOW_CONTENT_CAPTURE	https://developer.android.com/reference/android/os/UserManager#DISALLOW_CONTENT_CAPTURE
Android Enterprise - What is new in Android 17 / agentic automation controls	https://developer.android.com/work/versions/android-17
Google Play - July 15, 2026 policy announcement / third-party AI data responsibility	https://support.google.com/googleplay/android-developer/answer/17134731?hl=en

Revalidation note

Android APIs, Google Play policy and enterprise controls change. Re-check current official documentation at the start of P6, again at P12, and immediately before Play Store/enterprise distribution submission.

Current planning handoff

The Live Assist shell, unified Assistant, Orchestrator, Knowledge Gateway and Evidence Fabric are now planned at architecture/build-plan level. The next subsystem recommended after this document is Communications + Contact Intelligence. Implementation should begin only through the typed contracts and safety gates defined here, not through direct model-to-database or model-to-device control.

15. Assistant Brain / Orchestrator build plan

The LeaseOS Assistant should be one user-facing assistant backed by a typed hub-and-spoke orchestrator. Specialist capabilities do not receive independent authority. They expose bounded read/prepare/execute functions to the orchestrator, while deterministic LeaseOS engines remain authoritative for dispatch readiness, HOS evaluation, routing constraints, maintenance release and other domains that already have rule engines.

15.1 Target subsystem

```

AssistantOrchestrator
├── ContextResolver
├── IntentRouter
├── CapabilityRegistry
├── WorkflowCompiler
├── TaskGraphEngine
├── EvidenceResolver
├── ConflictResolver
├── ModelRouter
├── AuthorityGateway
├── ConfirmationEngine
├── TaskInbox
└── AuditReceiptWriter

```

Design rule: the model may propose a workflow, but only registered typed capabilities and actions may survive compilation and reach an executor.

15.2 Canonical context envelope

```

AssistantContext {
  tenantId, userId, authenticatedRoles[], temporaryDelegations[],
  deviceId, deviceTrust, managedDevice,
  operatingState: PARKED | MOVING | EMERGENCY,
  connectivity: ONLINE | DEGRADED | OFFLINE,
  activeJobId?, activeTripId?, activeLoadId?,
  unitId?, trailerId?, customerId?, location?,
  liveAssistSessionId?,
  availableCapabilities[], restrictedCapabilities[],
  observedAt
}

```

- Unknown context is represented explicitly as unknown; the model must not infer a unit, customer, job or jurisdiction simply because one is likely.
- The context snapshot is attached to each workflow and receipt so later review can reproduce the authority and state that existed when the request ran.
- Vehicle motion state and device management state are first-class policy inputs, not UI-only flags.

15.3 Multi-label intent and capability routing

Intent routing must allow one request to invoke several bounded capabilities. Example: "I unloaded at the disposal site, attach this ticket, finish the paperwork and tell me if I can go home" can require disposal, document capture, paperwork, HOS and routing capabilities in one task graph.

Layer	Responsibility	Must not do
IntentRouter	Classify primary intent plus required secondary capabilities.	Execute operational writes.
CapabilityRegistry	Declare inputs, outputs, scopes, risk, offline support, confirmation and	Grant authority not held by the current

	evidence requirements.	user.
WorkflowCompiler	Convert model plan into allowlisted typed steps.	Pass arbitrary model-generated commands to executors.
TaskGraphEngine	Track dependencies, resumability, waiting states and cross-role handoffs.	Hide blocked or incomplete steps.
AuthorityGateway	Evaluate tenant, role, job/unit scope, driving state, evidence and confirmation.	Accept model confidence as authorization.

15.4 Capability manifest contract

```
CapabilityManifest {
  capabilityId, version,
  inputSchema, outputSchema,
  readScopes[], writeScopes[],
  allowedRoles[], allowedDrivingStates[],
  riskLevel, confirmationPolicy,
  requiredEvidence[], authoritativeEngines[],
  offlinePolicy, reversibility,
  auditEvents[]
}
```

- Capabilities are functions, not personas. Examples: paperwork.complete, hos.evaluate, route.calculate, maintenance.create_work_order, billing.reconcile_job, knowledge.search_regulatory.
- Deterministic engines outrank general model opinion. The model gathers inputs, invokes the engine, explains the result and helps resolve deficiencies.
- A capability can be disabled by tenant policy or managed-device policy, but tenant policy cannot weaken the LeaseOS non-bypassable safety floor.

15.5 Workflow plan and task graph

```
WorkflowPlan {
  workflowId, requestId, contextSnapshotHash,
  steps: [{ actionId, inputs, dependsOn[], riskLevel, status }],
  requiredConfirmations[],
  unresolvedQuestions[],
  createdAt, expiresAt?
}
```

- Every step is idempotent where possible and includes an idempotency key for external writes.
- Long-running workflows persist as tasks so they can resume after loss of connectivity, a shift change, a mechanic handoff or a delayed signature.
- Cross-role handoffs become explicit tasks: driver defect -> mechanic work order -> authenticated release -> dispatch readiness re-evaluation.

15.6 Conflict resolution and evidence precedence

- Authority precedence is domain-specific. A current road closure may outrank a company preferred route; an OEM service specification may outrank generic mechanic notes; approved customer billing rules may govern customer-specific paperwork.
- When authoritative sources genuinely conflict, return CONFLICT and create a review task. Never average, vote or let the language model silently pick one.
- Each answer separates FACT, CALCULATION, DETERMINISTIC VERDICT, INFERENCE and RECOMMENDATION so the user can see what LeaseOS knows versus what it is reasoning about.

15.7 Assistant task inbox

ASSISTANT TASKS

READY Job 2817 billing package ready for review

NEEDS INPUT Job 2821 disposal ticket missing

BLOCKED Unit 214 unresolved brake defect

WAITING Permit verification waiting for connectivity

- Tasks can be assigned to driver, dispatcher, mechanic, office, safety, manager or a specific employee.
- Task state must be queryable independently of chat history; chats are an interface, not the workflow database.

15.8 Failure containment and concurrency

- Failure of one capability does not invalidate unrelated LeaseOS functions. Cloud-model failure must leave deterministic/offline functions available.
- Use recordVersion/expectedVersion and transactional checks on consequential updates. If a record changes during review, refresh rather than overwrite.
- External writes use idempotency keys and replay protection so duplicate model/tool calls cannot create duplicate invoices, work orders, tickets or communications.

15.9 Assistant audit receipt

```
AssistantReceipt {
  receiptId, requestSummary, requestedBy, actingRole,
  workflowId, capabilitiesUsed[], evidenceRefs[],
  deterministicVerdicts[], aiInferences[],
  confirmations[], actions[], failures[],
  finalState, contextSnapshotHash,
  startedAt, completedAt
}
```

- Receipts are designed for debugging, dispute review, compliance review and support; they do not need to retain raw chain-of-thought or transient screenshots.

15.10 Orchestrator implementation checkpoints

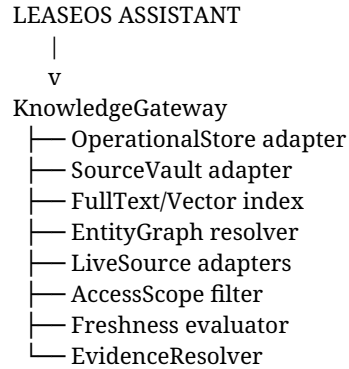
Checkpoint	Build	Pass condition
AO-1	AssistantContext + context snapshot persistence	Every request receives tenant/user/device/role/driving/connectivity/job context with explicit unknowns.
AO-2	CapabilityRegistry + schema validation	Unregistered capability IDs and invalid arguments fail closed.
AO-3	WorkflowCompiler	Model-generated plan can only compile to allowlisted typed actions.
AO-4	AuthorityGateway integration	Role/tenant/driving-state/risk/confirmation matrix covered by automated tests.
AO-5	TaskGraphEngine + TaskInbox	Workflow can pause, persist, resume and hand off across roles without chat state.
AO-6	Deterministic engine adapters	HOS/dispatch/routing/maintenance boundaries are invoked rather than

		duplicated in prompts.
AO-7	ConflictResolver + explainability payload	Conflicts are surfaced; Why-view can reproduce evidence and engine verdict.
AO-8	AuditReceiptWriter + replay/idempotency	Every consequential workflow yields a receipt and duplicate execution is prevented.

16. Knowledge / RAG Gateway build plan

Retrieval-augmented generation helps LeaseOS locate relevant knowledge, but the search index is not itself the source of truth. Original source records, verified operational data and deterministic rule engines remain authoritative. Search chunks and embeddings are replaceable indexes that point back to registered sources.

16.1 Knowledge Gateway architecture



16.2 Source authority classes

Class	Meaning	Examples / use
A0	LeaseOS deterministic result	HOS engine, dispatch readiness, route constraint verdict, billing reconciliation.
A1	Authoritative regulatory	Government/regulator/legal publication; jurisdiction and effective date required.
A2	OEM / technical authority	Manufacturer manual, service bulletin, approved technical specification.
A3	Approved company/customer policy	SOP, customer procedure, rate/billing/document rule.
A4	Verified LeaseOS business record	Inspection, signed field ticket, work order, GPS event, unit record.
A5	Live operational source	511, weather, facility, telematics; freshness-sensitive.
A6	Historical/user-generated	Notes, prior jobs, driver/mechanic observations; useful but not automatically current.
A7	General AI knowledge	Explanation only; never authoritative operational evidence.

16.3 Knowledge source registry

```
KnowledgeSource {
  sourceId, tenantId?, sourceType, title, publisher, authority,
  jurisdiction?, customerId?, equipmentMake?, equipmentModel?,
  sourceUrl?, originalFileId?, apiConnectorId?,
  version, revision?, publishedAt?, effectiveFrom?, effectiveUntil?,
  retrievedAt, verifiedAt?, verifiedBy?,
  verificationStatus: UNVERIFIED | REVIEWED | VERIFIED | SUPERSEDED | REVOKED,
  confidentiality: PUBLIC | COMPANY | CUSTOMER | RESTRICTED,
  licenseClass, redistributionAllowed, refreshPolicy, contentHash
}
```

- A same-name file with a different content hash is a new source revision, not a silent replacement.
- Regulatory sources require jurisdiction, version/effective dates and explicit verification before consequential use.
- Historical job facts stay historical until the current job explicitly confirms them.

16.4 Ingestion and indexing pipeline

```
Upload / API / approved website
|
source registration + integrity hash
|
file validation / licence-use check
|
text/table/section extraction
|
classification + entity linking
|
review if required
|
publish source revision
├── full-text index
├── embeddings
└── structured facts
```

- Original documents remain in the Source Vault. Embeddings and chunks may be rebuilt without changing source identity.
- Retrieved document content is untrusted data. It cannot change system policy, permissions, tool scopes or action contracts; this is the core prompt-injection boundary.

16.5 Hybrid retrieval and entity scoping

- Combine full-text/keyword, semantic retrieval, structured database queries, entity relationships, authority, jurisdiction, customer, equipment and effective-date filters.
- Equipment queries resolve unit -> VIN/configuration/components -> applicable documents before retrieval. Do not retrieve a generic manual first and guess applicability afterward.
- Customer queries are tenant/customer scoped before retrieval so one customer's procedures, contacts or rates cannot leak into another customer's context.
- Authorization is applied before search. The system never retrieves across unauthorized corpora and then tries to hide the result afterward.

16.6 Knowledge answer contract

```
KnowledgeAnswer {
  answer, statements[], evidence[],
  authorityLevel, confidence, freshAsOf, staleSources[],
  conflicts[], unknowns[], inferenceUsed,
  applicableJurisdiction?, applicableEquipment?, applicableCustomer?
}
```

- Consequential answers support a Why view with source, revision, effective date, freshness, evidence and any inference used.
- If live data is stale or unavailable, show UNKNOWN/STALE rather than presenting cached information as current.

16.7 Offline knowledge packs

```
KnowledgePack {
  packId, tenantId, userId/deviceId,
  sourceRevisionIds[],
  createdAt, expiresAt,
  hash, signature
}
```

- Preload only job-relevant customer procedures, ERP/ERAP information, emergency contacts, permits, unit manuals, offline map/routing data and applicable rules.
- Signed pack metadata lets LeaseOS later show exactly which source revisions were present on a field device at a given time.

16.8 Governance and refresh center

```
KNOWLEDGE GOVERNANCE
NEEDS REVIEW  source requires human review
EXPIRING     policy/source nearing review/effective end
SOURCE CHANGED upstream authoritative source changed
CONFLICTS    two active sources disagree
UNVERIFIED   imported but not verified
SUPERSEDED   retained for historical reproducibility
```

- Live adapters declare authority, authentication, refresh interval, fields, retention, failure state and rate limits.
- Arbitrary web pages may assist research but cannot automatically become authoritative operational evidence.

16.9 Recommended logical stores

Store	Contains	Rule
Operational Store	Jobs, trips, units, HOS, tickets, billing, maintenance, users.	Transactional source for current LeaseOS business state.
Source Vault	Original PDFs/manuals/SOPs/regulatory snapshots.	Immutable/versioned originals with hashes and retention.
Search Index	Full-text, embeddings, entity index.	Rebuildable; never authoritative on its own.
Knowledge Governance	Source metadata, licence, authority, verification, effective dates, reviews.	Controls whether/where a source may be used.

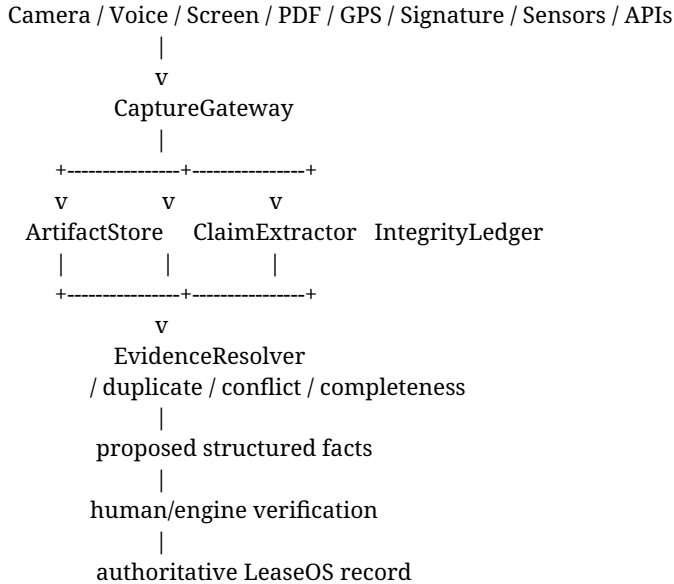
16.10 Knowledge implementation checkpoints

Checkpoint	Build	Pass condition
KG-1	KnowledgeSource registry + revisions	Every source has authority, revision/hash, scope, verification and retention metadata.
KG-2	Source Vault + extraction pipeline	Originals preserved; searchable derivatives link back to exact source revision.
KG-3	AccessScope filter	Tenant/role/customer restrictions apply before retrieval; cross-tenant tests pass.
KG-4	Hybrid retriever	Structured + keyword + semantic + entity/date/jurisdiction filtering works together.
KG-5	Prompt-injection boundary	Retrieved content cannot request tools, alter permissions or override system/action policy.
KG-6	Freshness/conflict engine	Stale/changed/conflicting sources are labeled and consequential answers fail closed where required.
KG-7	KnowledgeAnswer + Why view	Every important statement can be traced to evidence/source or labeled inference.
KG-8	Offline signed knowledge packs	Pack is scoped, encrypted/signed, expires correctly and supports source-revision audit.
KG-9	Governance review center	Admins can review, supersede, revoke and resolve source conflicts without deleting history.

17. Multimodal Evidence Fabric build plan

The Evidence Fabric is the common ingestion, integrity and claim-resolution service for camera images, paper tickets, PDFs, screen observations, voice, GPS, signatures, sensors, telematics and external API records. Evidence may propose a fact; it does not silently become an authoritative job, billing, HOS, safety or maintenance record.

17.1 Evidence architecture



17.2 Canonical evidence contracts

```

EvidenceArtifact {
  evidenceId, tenantId, capturedBy, deviceId,
  evidenceType, mimeType?, capturedAt, receivedAt,
  latitude?, longitude?, accuracyMeters?,
  jobId?, tripId?, loadId?, unitId?, trailerId?, employeeId?, customerId?, facilityId?,
  originalHash, storageReference, sourceApp?, captureMethod,
  integrityStatus, retentionClass, parentEvidenceId?
}
  
```

```

EvidenceClaim {
  claimId, evidenceId, field, proposedValue, unit?,
  extractionMethod, precision?, confidence, status,
  corroboratingEvidenceIds[], contradictoryEvidenceIds[]
}
  
```

- Artifact identity and claim identity are separate. One ticket image may support several field-level claims with different confidence.
- Corrections never overwrite extraction history; store original proposal, correction, actor, reason and original evidence reference.

17.3 Evidence classes and quality gates

Class	Examples	Quality / trust checks
DOCUMENT	Field ticket, disposal ticket, scale ticket, permit, inspection, receipt.	Readability, page/corner completeness, identifiers, signatures, duplicate detection.
PHOTO	Defect, load, equipment, site condition,	Blur/glare, timestamp, contextual linkage,

	meter.	optional annotation.
VOICE	Driver statement, mechanic note.	Transcription confidence, precision such as approximate vs exact, speaker/session context.
SCREEN_CAPTURE	Live Assist observation.	Source app, session consent, transient frame handling, prefer official API if available.
GPS	Arrival/departure/location.	Accuracy radius, provider, timestamp; uncertain geofence -> pending.
SENSOR/TELEMATICS	Temperature, pressure, tank level, axle weight, fault code.	Sensor identity, unit, calibration/health, sample time, derived vs measured.
SIGNATURE	Customer/company/driver signoff.	Payload hash, signer identity/role, consent text version, time/device.
API_RECORD	Facility/scale/telematics/511.	Connector authority, freshness, source identity, failure state.

17.4 Field-level extraction and corroboration

- Confidence is per field, not one document-wide score. Ticket number may be 99% while signature presence is 61%; only uncertain fields should require review.
- Corroboration increases support but does not justify averaging unlike measurements into a fabricated value. Domain rules choose the canonical evidence source.
- Billing-sensitive conflicts create exceptions instead of silent selection. Example: driver says 18.4 m3 while ticket/API show about 16.8 m3 -> review required.

17.5 Camera / OCR workflow

Capture document

- > document edge/orientation/quality check
- > OCR/barcode/QR extraction
- > field-level claims + confidence
- > job/load/facility matching
- > duplicate + wrong-job checks
- > targeted human review
- > attach evidence + confirmed fields

- If required fields are unreadable or a required signature area is cut off, recommend/require a retake according to workflow policy before billing closeout.
- Use perceptual hash plus extracted identifiers and date/facility/quantity combinations to detect duplicate ticket photos.

17.6 Mechanic evidence workflow

DEFECT

BEFORE photo/measurement

- > work order / parts / technician notes
- > AFTER photo / test result
- > authorized return-to-service release

- AI visual observations remain observations/inferences until a mechanic verifies them.

- Return-to-service is an authenticated authority boundary and cannot be replaced by AI confidence or a clean-looking photo.

17.7 Sensor and derived-measurement rules

- Store sensor identity, engineering unit, timestamp, calibration status and health state with every material measurement used operationally.
- Derived values retain their formula/input chain. Example: tank level + geometry -> estimated volume must remain CALCULATED, not MEASURED.
- Low-health or overdue-calibration sensors can remain supporting evidence but should not silently become billing or safety authority.

17.8 GPS, temporal and location reasoning

- GPS claims include accuracy; poor accuracy yields suggested/pending arrival rather than confirmed arrival.
- Temporal checks flag impossible or suspicious ordering, such as a customer signature predating job start, without making accusations.
- Capture location is not proof of event location when evidence may have been photographed later; the system records both and explains the distinction.

17.9 Signature and evidence bundle integrity

```
SignatureRecord {
  signer, signerRole, signedObjectType, signedObjectId,
  payloadHash, signedAt, deviceId, signatureMethod,
  location?, consentTextVersion
}
```

- A signature binds to an exact payload hash. Post-signature changes require an amendment/new signature path rather than mutating the signed record.
- Important job, maintenance and billing packets expose an evidence bundle containing source artifacts, confirmations, signatures, engine results and audit receipts.
- Hash/integrity mismatch is an explicit failure state; originals are never silently replaced.

17.10 Completeness engine

```
EvidenceRequirementSet
  workflowType: DISPOSAL_CLOSEOUT
  required: tracking, facility, quantity, ticket, timestamps, operator, material, acknowledgment
```

```
Result
  SATISFIED: 6/7
  MISSING: facility acknowledgment
```

- The AI Secretary asks only for the minimum missing evidence because completeness is deterministic, not prompt-dependent.
- The Assistant cannot manufacture a missing signature, facility acknowledgment or measurement; it can request it, create a task or route an authorized exception process.

17.11 Privacy and retention

- Whole-screen frames and OCR intermediates are transient by default; only scoped evidence required for the workflow becomes a durable business artifact.
- Unrelated personal content observed during a Live Assist session is not automatically persisted or indexed.
- Business documents, signed tickets and repair evidence follow the applicable business/compliance retention class; intermediate model reasoning is minimized.

17.12 Evidence Fabric implementation checkpoints

Checkpoint	Build	Pass condition
EF-1	EvidenceArtifact + immutable storage refs/hash	Every artifact has identity, scope, source, time and integrity metadata.
EF-2	EvidenceClaim + field-level provenance/confidence	Extracted facts remain proposals until confirmation/engine verification.
EF-3	Document quality + OCR pipeline	Poor/partial evidence is detected; field-level review works.
EF-4	Duplicate/wrong-job detector	Duplicate evidence and mismatched job/load identifiers create review, not duplicate writes.
EF-5	Voice/GPS/screen adapters	Approximate voice and uncertain GPS preserve precision/accuracy; screen frames remain scoped/transient.
EF-6	Sensor/telematics adapter	Measurements preserve calibration/health and derived-vs-measured distinction.
EF-7	Signature payload binding	Signature verifies against exact payload; altered payload cannot pass verification.
EF-8	Conflict/completeness engines	Contradictions create exceptions and missing evidence produces minimum-question tasks.
EF-9	Evidence bundles + integrity ledger	Consequential record can enumerate evidence/confirmations/receipts and detect tampering.
EF-10	Retention/privacy enforcement	Transient frames/intermediates expire; durable evidence follows declared retention class.

18. Integrated implementation sequence and checkpoint gates

Version 2.0 does not replace the phases in Sections 9-13. It makes the Assistant, knowledge and evidence work explicit and orders them so authority, provenance and testability exist before more powerful AI or screen-control behavior is enabled.

Order	Work package	Key dependencies	Exit gate
1	Core contracts: Provenance, AssistantContext, EvidenceArtifact/Claim, CapabilityManifest, ActionDefinition	Existing tenant/roles/audit schema	Schema + type tests green; no duplicate parallel provenance concepts.
2	AuthorityGateway + ConfirmationEngine + action registry	Core contracts	Fail-closed authorization matrix and L4 confirmation tests green.
3	Evidence Fabric foundation	Object storage/local encrypted store, provenance	Capture -> claim -> confirm pipeline works offline and online.
4	Knowledge source registry + Source Vault + scoped retriever	Tenant/customer/equipment model	Source revision, verification, ACL and prompt-injection tests green.
5	Orchestrator AO-1..AO-4	Authority + knowledge/evidence read APIs	Typed workflows compile; invalid model actions cannot execute.
6	AI Secretary form engine integration	Orchestrator + Evidence Fabric	Voice/OCR fields preserve precision/source/confidence and require read-back as configured.
7	TaskGraph + Assistant Task Inbox	Orchestrator	Pause/resume/handoff works across driver/mechanic/office roles.
8	Live Assist in-app, then whole-screen read-only	Android permission shell + evidence adapters	Consent/revoke/protected-screen/transient-retention tests green.
9	Deterministic Android/third-party action adapters	AuthorityGateway + registered action vocabulary	No unrestricted model tap loop; idempotency and rollback policy tested.
10	Model router + offline packs	Knowledge/Evidence/Orchestrator	Offline degradation labels stale/unknown and queues safe work.
11	Mechanic, billing, dispatch, HOS, routing capability expansion	Existing deterministic engines	Each capability proves engine precedence and role isolation.
12	Managed fleet/MDM policy	Android enterprise	Managed restrictions can

	mode	deployment	reduce capability but never expand user/AI authority.
13	Proactive tasking + notifications	TaskGraph and evidence completeness	Interruptions obey driving state and company policy.
14	Pilot + production gates	All above	Cross-tenant, replay, offline, privacy, authority and human-release gates green.

18.1 Mandatory cross-cutting tests

- Tenant isolation across operational records, search indexes, evidence artifacts, task queues and audit receipts.
- Role/temporary-delegation matrix for every typed action and capability.
- Parked/moving/emergency state enforcement at gateway level even when a model requests a prohibited action.
- Unknown/stale/conflicting source behavior; unknown is never converted to safe/clear/verified.
- Field-level precision and provenance for voice/OCR; approximate values stay approximate until confirmed.
- Mechanic release and dispatch readiness cannot be bypassed by AI recommendations.
- Idempotency/replay protection for work orders, invoices, ticket submission, messages and external actions.
- Optimistic concurrency and transactional conflict handling for simultaneous office/field updates.
- Prompt-injection tests against uploaded manuals, PDFs, web snapshots and OCR text.
- Live Assist capture lifecycle: denied, started, backgrounded, rotated, revoked, protected screen, process death and reboot.
- Retention and redaction: transient screen/audio/OCR intermediates expire and secrets are absent from audit logs.
- Source/version reproducibility: important verdicts can show which source revision and evidence were used.

18.2 Production-ready definition for this subsystem set

- The Assistant can explain and prepare work without bypassing existing dispatch, HOS, routing, billing, signature or mechanic-release authority boundaries.
- Every consequential workflow has a typed plan, explicit evidence, confirmation record where required, and an AssistantReceipt.
- Knowledge retrieval is tenant/role/customer/equipment scoped before search and can identify stale, superseded, conflicting or unverified sources.
- Evidence ingestion preserves originals, field-level claims and amendment history; duplicate/wrong-job/conflicting evidence is surfaced instead of silently merged.
- Offline behavior degrades safely and synchronizes without duplicating or overwriting newer authoritative state.
- Live Assist is visibly active, revocable, transient by default, and incapable of turning general AI reasoning into unrestricted device control.

19. Current planning boundary and next subsystem

The current planning tranche is complete through the Live Assist interaction layer, unified AI Secretary, Assistant Orchestrator, Knowledge/RAG Gateway and Multimodal Evidence Fabric. This is enough architecture to begin a disciplined implementation plan and checkpoint build without inventing authority, storage or evidence behavior during coding.

The next recommended design subsystem is Communications + Contact Intelligence: company personnel, drivers, dispatchers, consultants, clients, disposal facilities, medics, mechanics, ERAP/emergency contacts and historical job contacts, with role resolution, freshness, consent, escalation chains and audited communications.

This document intentionally stops before designing that communications/contact subsystem so the current build plan remains a coherent checkpoint rather than silently expanding scope.

19.1 Project grounding used for Version 2.0

- LEASEOS_BUILD_PLAN-3.md: one schema-driven AI Secretary engine; schema-constrained extraction; per-field value/precision/source/confidence; minimum missing questions; mandatory read-back; commit with provenance; confirmation as a core safety property.
- leaseos-integrated-ops.diff: authenticated mechanic return-to-service boundary; unified dispatch readiness engine; source-versioned HOS boundary; evidence-linked billing; immutable amendment history; provenance-aware map/regulatory sources; AI Secretary uncertainty + human confirmation; production tests for role authorization, dispatch rejection, sync conflicts and cross-customer isolation.
- Existing Version 1.0 Live Assist plan: BYOD + managed Android modes, hybrid local-first processing, parked/moving/emergency action policies, typed action vocabulary, offline behavior, privacy and audit foundations.